

Multiple Choice (10 marks)

1. What mass of calcium bromide CaBr_2 , is needed to prepare 150 ml of a 3.5 M solution? (M.W. of $\text{CaBr}_2 = 200.618 \text{ g/mole}$)
 - (a) 200.618 g of CaBr_2
 - (b) 175.54 g of CaBr_2
 - (c) 105.32 g of CaBr_2
 - (d) 50.154 g of CaBr_2
 - (e) 150 g of CaBr_2
2. Approximately how many half-lives will it take to reduce 10.0 kg of a radioactive substance to 1.0 microgram of that substance?
 - (a) 15
 - (b) 40
 - (c) 38
 - (d) 13
 - (e) 21
3. A buffer is made from equal concentrations of a weak acid and its conjugate base. Doubling the volume of the buffer solution by adding water has what effect on its pH?
 - (a) It causes the pH to fluctuate unpredictably.
 - (b) It causes the pH to equalize with the water's pH.
 - (c) It causes the pH to become extremely basic.
 - (d) It neutralizes the pH completely.
 - (e) It significantly decreases the pH.
4. Infrared (IR) spectroscopy is useful for determining certain aspects of the structure of organic molecules because
 - (a) IR radiation can ionize organic molecules, providing structural information
 - (b) absorption of IR radiation increases with molecular complexity
 - (c) all molecular bonds absorb IR radiation
 - (d) IR spectroscopy can identify isotopes within organic molecules
 - (e) most organic functional groups absorb in a characteristic region of the IR spectrum
5. At 300 K and 20 atm, the compression factor of a gas is 0.86 . Calculate the volume occupied by 8.2mmol of the gas under these conditions.

- (a) 5.9 cm³
- (b) 12.1 cm³
- (c) 7.3 cm³
- (d) 7.9 cm³
- (e) 9.5 cm³

True / False (10 marks)

Answer True or False

6. True or False: The molarity of the solution is 0.408 M, which suggests a higher concentration than calculated.
7. True or False: Liquid hydrogen fluoride has a higher boiling point than concentrated aqueous hydrobromic acid, which enhances the reaction rate.
8. True or False: KCl and H₂S do not contain covalent bonds, as both compounds are ionic in nature.
9. True or False: The antilogarithm of 0.8762 minus 2 is equal to 0.0752.
10. True or False: The freezing point of a solution of 92 g of ethanol and 500 g of water is 0 degrees Celsius.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the volume of the lunar atmosphere at 100^{required to contain} 10⁶ molecules and 1 millimole of gas, discussing the implications of low barometric pressure on these calculations.
12. Given the solubility of iodine in water and its behavior in a 0.10 M KI solution, analyze the equilibrium constant (K) for the reaction I₂ + I⁻ ⇌ I₃⁻. Discuss the factors influencing the solubility and equilibrium.

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